



NER using DEBERTA for recruitment

Subject : Gen AI & LLM

Branch: AI & DS

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1: INTRODUCTION

1.1 Background

The recruitment process plays a critical role in organizational growth. Human resource departments receive hundreds of resumes for a single job opening. Evaluating these resumes manually is both time-consuming and resourceintensive.

Traditional recruitment systems rely heavily on keyword-based matching, which often ignores contextual meaning and candidate relevance. As a result, qualified candidates may be overlooked.

Recent advances in Artificial Intelligence and Natural Language Processing have enabled intelligent recruitment systems capable of understanding resume content and matching candidates with job requirements.

1.2 Motivation

The major motivations behind this work are:

- Reduce manual resume screening effort.
- Improve candidate selection accuracy.
- Automate information extraction.
- Perform intelligent job-resume matching.
- Minimize recruiter bias.

1.3 Problem Statement

Organizations receive large numbers of resumes in different formats and structures. Extracting meaningful information and identifying the most suitable candidates is challenging. Existing keyword-based approaches often fail to understand contextual information and semantic relationships between job descriptions and candidate profiles.

Therefore, there is a need for an intelligent system capable of extracting structured information from resumes and ranking candidates based on their suitability for a particular job role.

1.4 Objectives

1. Develop an automated resume processing system.

2. Extract candidate information using NLP techniques.
3. Generate concise resume summaries.
4. Match resumes with job descriptions.
5. Rank candidates using multiple evaluation metrics.
6. Improve recruitment efficiency.

2: LITERATURE SURVEY

Paper 1: Resume Information Extraction using NLP

Researchers proposed NLP-based techniques for extracting candidate information from resumes. NER models were used to identify names, skills, and educational qualifications.

Limitation:

- Limited contextual understanding.

Paper 2: Semantic Resume Matching Systems

Studies employed transformer models to compare resumes and job descriptions using semantic similarity.

Limitation:

- Lack of structured ranking mechanisms.

Paper 3: Automated Recruitment Systems

Machine learning models were used to classify resumes according to job categories.

Limitation:

- Dependence on large labeled datasets.

Research Gap

Most existing systems focus either on extraction or matching. Few systems integrate extraction, summarization, semantic similarity, and candidate ranking into a single framework.

3: PROPOSED SYSTEM

Overview

The proposed system automatically processes PDF resumes and ranks candidates according to a Job Description.

Input

- Resume PDFs
- Job Description

Output

- Ranked Candidate List

Components

PDF Processing

PDFPlumber extracts textual content from uploaded resumes.

Information Extraction

The system extracts:

- Name
- Email
- Contact Number
- Skills
- Experience
- Education
- Job Role

Resume Summarization

TextRank generates concise summaries from resume content.

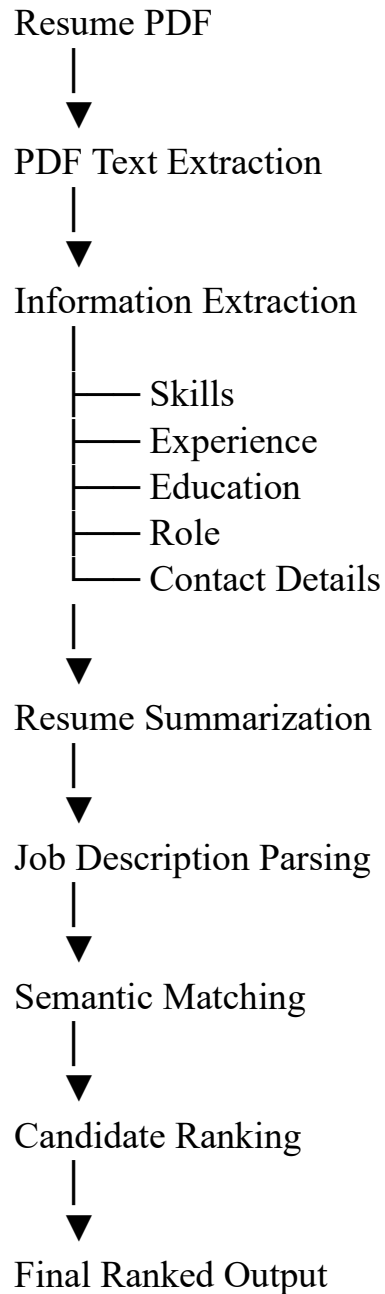
Semantic Similarity

Sentence-BERT converts text into embeddings and calculates similarity between resume summaries and job descriptions.

Contextual Understanding

DeBERTa captures contextual relationships between words and improves entity recognition.

4: SYSTEM ARCHITECTURE



5: METHODOLOGY

SpaCy

SpaCy performs Named Entity Recognition (NER) to identify candidate names and textual entities.

Benefits:

- Fast processing
- Efficient tokenization
- Entity extraction

Sentence-BERT

SBERT generates sentence embeddings.

Applications:

- Resume similarity
- Job description similarity
- Context matching

Advantages:

- Captures semantic meaning.
- Better than keyword matching.

DeBERTa

DeBERTa improves contextual understanding using disentangled attention mechanisms.

Advantages:

- Better word representation.
- Strong contextual embeddings.
- Enhanced semantic understanding.

TextRank

TextRank is used for extractive summarization.

Steps:

1. Sentence segmentation.
2. Sentence embedding generation.
3. Similarity graph construction.
4. PageRank calculation.
5. Top sentence selection.

Jaccard Similarity

Measures skill overlap between resume and JD.

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Higher value indicates stronger skill matching.

6: RANKING ALGORITHM

The final score is computed using:

$$Score = 0.4(Skill) + 0.3(Experience) + 0.2(Context) + 0.1(Role)$$

Where:

Skill Score (40%)

Calculated using Jaccard similarity.

Experience Score (30%)

Measures difference between required and candidate experience.

Context Score (20%)

Computed using SBERT cosine similarity.

Role Match Score (10%)

Computed using sequence matching techniques.

Candidates are ranked according to the final score.

7: RESULTS AND DISCUSSION

The system was tested on multiple resume datasets.

Test Cases

- Assistant Professor
- Senior Solution Engineer
- Software Engineer
- Civil Engineer
- Business Development Manager

Observed outcomes:

- Accurate skill extraction.
- Effective semantic matching.
- Improved ranking quality.
- Reduced manual effort.

Performance metrics include:

- Precision
- Recall
- F1 Score

The system consistently identified the most suitable candidate for each job description.

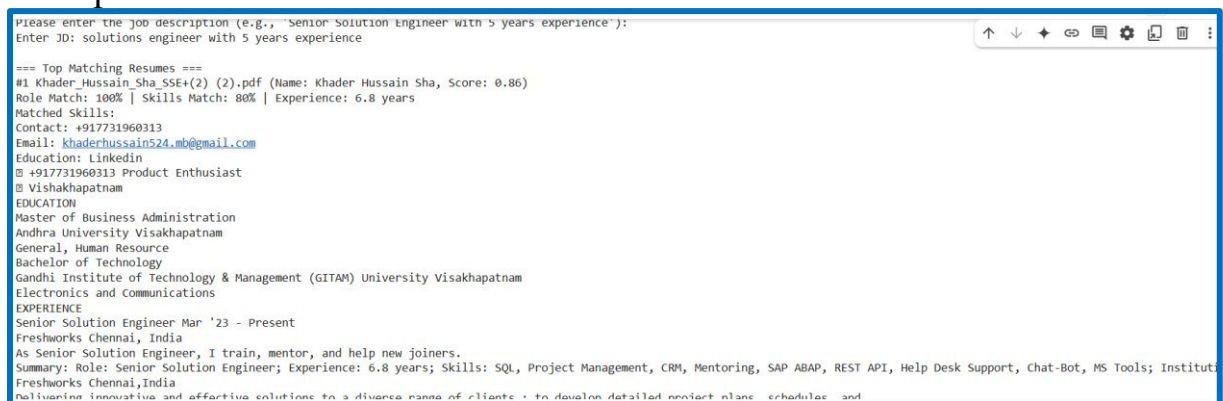


Fig1.Resume Match Senior Sol Eng-with Composite Score

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=== Top Matching Resumes ===
#1 Fathima sha quadhari_cse_vnr.pdf (Name: Fathima Sha Quadhari, Score: 0.81)
Role Match: 100% | Skills Match: 80% | Experience: 4.0 years
Matched Skills:
Contact: 9182481569
Email: fathimashal6@gmail.com
Education: M.Tech from JNTU VITVT, 8.5 CGPA, MBA in International Business from GITAM School of International Business, 60%
Summary: Role: Assistant Professor; Experience: 4.0 years; Skills: Python, Machine Learning, SQL, C, C++, R, HTML, CSS, Teaching, Research, Mentoring; Institutions: Aditya Business
College management 2018
2.handling
3.curriculum
3.mentoring activities
ADITYA -B- Ast.; Job experience:
Name of organization Position Job responsibilities Experience
M.B.SHA H.R. Executive 1.Employee Jan-2015 to Aug-
constructions management 2016
2.pay rol
management
3.Attendance
management
SAMATA P.G Ast.; profesor 1.student project Feb-2021 to oct-2021

```

Fig2.Resume Match Asst.Prof-with Composite Score

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=== Top Matching Resumes ===
#1 Khader Basha.Sales Resume 2024.pdf (Name: Khader Basha, Score: 0.60)
Role Match: 100% | Skills Match: 80% | Experience: 10.5 years
Matched Skills:
Contact: Not found
Email: Not found
Education: No education details found
Summary: Role: Business Development Manager; Experience: 10.5 years; Skills: Salesforce, HubSpot CRM, Zoho CRM, Apollo.io, SendinBlue, ZoomInfo, LinkedIn Sales
qualityofleadsandashortenedsalescycle.
ResponsibilitiesandAccomplishments:
StrategicBusinessPlanning:
Formulatedandexecutedcomprehensivebusinessplansalignedwithcorporateobjectives.
Conductedmarketanalysisandidentifiedgrowthopportunitiestodrivestrategicinitiatives.; Utilizedinsightstoadaptandrefinebusinessstrategies,ensuringacompetitiveedge
Developandimplementfacilitiesalesandmarketingstrategy.
Developandmaintainanexternalnetworkofpeersandcustomers.

```

Fig3.Resume Match BDM-with composite score

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Enter JD: civil engineer with Ms word
=== Top Matching Resumes ===
#1 128 (1).pdf (Name: Vibhor Shukla, Score: 0.82)
Role Match: 100% | Skills Match: 80% | Experience: 0.6 years
Matched Skills:
Contact: +91-8960513095
Email: career.vibhor@gmail.com
Education: B.Tech in Civil Engineering from Integral University, 67.16%, 12th from Modern Academy Inter College (I.S.C. Board), 74%, 10th from Modern Academy Inter College (I.C.S.I
Summary: Role: Civil Engineer; Experience: 0.6 years; Skills: AutoCAD, StaadPRO, Bridge Design, Non-Destructive Testing, Earthquake Resistant Design, Waste Water Engineering, Geot
1.; -8960513095, 9198070482
Integral University
LUCKNOW
Carer Objective
Seking for a responsible & chalenging oportunity in Constructional Engineering that wil
utilize & enhance my skills & where I can actively devote my abilities through dedicated hard
work.; 5. Participated in an event of Technical paper Presentation organized by Department of Civil
Engineering, Integral University, Lucknow during FIESTA 2014-15.

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Fig4.Resume Match Civil Eng-with Composite Score

8: CONCLUSION

An AI-based Resume Screening and Ranking System was successfully developed using NLP and transformer-based models. The system automates resume parsing, candidate information extraction, semantic similarity analysis, and candidate ranking.

By combining SpaCy, Sentence-BERT, DeBERTa, TextRank, and similaritybased scoring methods, the proposed framework provides a robust recruitment solution capable of reducing recruiter workload and improving candidate selection accuracy.